SCH3U **ATOMIC THEORY SELF STUDY**

Instructions:

- *Read p# ___________ in your textbook.*
- *On a separate piece of paper answer the following questions using full sentences.*

1. What contribution did the ancient Greeks make to the atomic theory?
2. In ancient times the equivalent of today's four states of matter (solids, liquids, gases, plasma) was…
3. List the postulates of John Dalton's "Atomic Theory".
4. Create a time line for the development of Atomic Theory from 1803 – 1920. (Include dates, scientist name & major contribution).
5. What are the three sub-atomic particles? What are their characteristics (include charges and weights)?
6. Describe Rutherford's gold foil experiment *and* sketch the experimental set-up.
7. Explain how the results of this experiment helped us 1) develop a greater understanding of the atom as a whole and 2) helped us led to the discovery of the nucleus.
8. What contribution(s) did Chadwick make to the development of atomic theory?
9. The atomic number is also known as the ______ number?
10. What information is given about an atom in the atomic number?
11. What information is given about in the mass number?
12. Define the term isotope?
13. List three isotopes of hydrogen?
14. What is meant by the abundance as it is related to isotopes?
15. Describe what is meant by radioactive isotopes?
16. Why is it that Dalton's postulate that "atoms are tiny indivisible particles" no longer valid?
17. What is a half-life?
18. What is carbon dating?